

TUHIN BANDOPADHYAY

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RESEARCH INTERESTS

Multiphase flows; Particle-laden turbulent flows; Turbulence modulation; Experimental fluid dynamics; Magnetic resonance velocimetry (MRV) and concentration; Urban flows; Turbomachinery; Flow control; Optical flow diagnostics

EDUCATION

University of Illinois, Urbana-Champaign

Champaign, IL

Doctoral student in Aerospace Engineering. GPA: 4.00/4.00

Sep. 2020 - Present

Indian Institute of Technology, Kharagpur

kharagpur, India

Dual Degree (B.Tech+M.Tech) in Aerospace Engineering. GPA: 9.50/10

Sep. 2015 - July 2020

AWARDS AND FELLOWSHIPS

Mavis Future Faculty Fellowship

2024-2025

The Grainger College of Engineering, UIUC

Graduate College Block Grant Fellowship

2024

Department of Aerospace Engineering, UIUC

Faculty Endowed Graduate Award

2024

Department of Aerospace Engineering, UIUC

Conference Travel Award

2022

Department of Aerospace Engineering, UIUC

Best Project Award for Bachelor and Master project

2020

Indian Institute of Technology, Kharagpur

ASME IGTI (International Gas Turbine Institute) Student Scholarship

2018-2019

The American Society of Mechanical Engineers (ASME)

Summer Research Fellowship

2018

Indian Academy of Science (IAS)

MCM Scholarship

2015-2018

Indian Institute of Technology, Kharagpur

JBNSTS Fellowship

2015-2016

Jagadish Bose National Science Talent Search

Academic Excellence Award

2015

Chief Minister, West Bengal, India

RESEARCH PROJECTS

Dynamics of inertial particle clusters in turbulent flows

Jan. 2021 - Present

Department of Aerospace Engineering, UIUC

Advisor: [Dr. Laura Villafañe](#)

- Have designed, developed, and characterized a particle-laden turbulent channel flow facility (the largest in the USA) to experimentally study the coupled interactions of inertial particles with turbulence.
- Experimentally investigating the coherent dynamics of the inertial particles for varying parameter space.

A novel tool for particle cluster identification and temporal tracking

Department of Aerospace Engineering, UIUC

Sep. 2020 - Sep. 2023
Advisor: [Dr. Laura Villafañe](#)

- A novel method has been developed based on the DBSCAN clustering algorithm to identify and temporally track particle clusters in particle-laden turbulent flows.
- The method is significantly cheaper computationally compared to the existing methods.
- Tracking of the identified clusters over space and time provides a comprehensive understanding of cluster dynamics and its correlation with underlying flow features.

Effect of wind direction on the ventilation dynamics of a model sports stadium

Collaborative project between West Point and UIUC

April 2023 - Present
Advisors: [Dr. Laura Villafañe](#), [Dr. Andrew J. Banko](#)

- Experimentally investigated scalar dispersion around a model stadium as a prototypical open-air structure.
- Novel medical imaging techniques are combined with laser-based diagnostics to obtain a comprehensive set of flow velocity and concentration fields.
- External and internal three-dimensional velocity fields are measured using magnetic resonance velocimetry, while time-resolved concentration fields at a central plane and outside the stadium are obtained using planar laser-induced fluorescence.

Three-dimensional velocity measurements in periodic flows via MRV

Department of Aerospace Engineering & Beckman Institute, UIUC

April 2021 - March 2022

Advisors: [Dr. Laura Villafañe](#), [Dr. Brad Sutton](#)

- The project is a part of the Magnetic Resonance Velocimetry (MRV) Challenge, 2021.
- Benchmarked the performance of the MRI machine, developed the post-processing tools, and established the optimal experimental framework to scan complex 3-dimensional periodic transient flows.

Characterization of the leading-edge separation bubble (Master's thesis)

Department of Aerospace Engineering, IIT Kharagpur

2018 - 2020
Advisor: [Dr. Chetan S. Mistry](#)

- Designed and developed an experimental setup (wind tunnel test section) to study separated boundary layer characteristics over the leading edge of a flat plate, with the provision to change the angle of incidence.
- Studied detailed flow physics inside leading-edge separation bubble for varying Reynolds numbers and angle of incidences using 2D PIV (particle image velocimetry).

Institute challenge research project: 'Airavat'

Department of Aerospace Engineering, IIT Kharagpur

2017 - 2020
Advisor: [Dr. Chetan S. Mistry](#)

- founding student leader and propulsion team head of the project. The aim is to develop a flying car propelled by green energy (electric).
- Secured funding of 1 million Rupees from Institute Innovation Grant through a competitive selection process.
- I was involved with the design of the propulsion system of the vehicle using axial flow ducted fans.

Instabilities in low Reynolds number round jets (Summer internship)

Department of Aerospace Engineering, IIT Kanpur

May 2018 - July 2018
Advisor: [Dr. Debopam Das](#)

- Experimentally investigated the behavior of a non-buoyant circular water jet discharged from a circular nozzle of diameter 1 mm for a Reynolds number range of 150 – 1000.
- Performed dye flow visualization and 2D PIV to understand the instability mechanisms before jet breakdown.

JOURNAL PUBLICATIONS

4. (*Under preparation*) **Tuhin Bandopadhyay**, Laura Villafañe. "Dynamics of inertial particle clusters in the core region of a turbulent duct flow." *International Journal of Multiphase Flow*, 2024.
3. Benson M., Banko A., Elkins C., An D.G., Song S., Bruschewski M., Grundmann S., **Bandopadhyay T.**, Sutton B., Villafane L., Han K., Hwang W., Eaton J.K. "MRV Challenge 2: Phase locked turbulent measurements in a

roughness array.” *Experiments in Fluids*, 64(28), 2023.

2. **Bandopadhyay, Tuhin**, Chetan S. Mistry. “Characterization of Leading-edge Laminar Separation Bubble for various Angles of Incidence and Reynolds Numbers.” *Journal of The Institution of Engineers (India): Series C*, 2022.
1. **Bandopadhyay, Tuhin**, Chetan S. Mistry. “Effects of Total Pressure Distribution on Performance of Small-Size Counter-Rotating Axial-Flow Fan Stage for Electrical Propulsion.” *ASME Open Journal of Engineering* 1, 2022.

CONFERENCE PUBLICATIONS/TALKS

7. **Bandopadhyay, Tuhin**, Laura Villafane. “Unraveling the persistence and dynamics of high-concentrated regions of inertial particles in turbulent flows.” *Bulletin of the American Physical Society*, 2023.
6. Villafane, L., **Bandopadhyay T.**, Sutton, B., Ellwein, B., Gehl, J., Iliff, S., ... & Banko, A. “Effect of Wind Direction on the Ventilation Dynamics of a Model Sports Stadium.” *Bulletin of the American Physical Society*, 2023.
5. **Bandopadhyay, Tuhin**, Laura Villafane. “Coherence of inertial particle clusters in the core region of a turbulent channel flow.” *Bulletin of the American Physical Society* 67, 2022.
4. **Bandopadhyay, Tuhin**, Laura Villafane. “Tracking particle clusters in turbulent channel flows via density-based clustering algorithms.” *APS Division of Fluid Dynamics Meeting Abstracts*, 2021.
3. Villafane, L., **Bandopadhyay, T.**, Tafti, S., & Sutton, B. “Three-dimensional velocity measurements in pulsatile flows via Magnetic Resonance Velocimetry, 2021 MRV Challenge.” *APS Division of Fluid Dynamics Meeting Abstracts*, 2021.
2. **Tuhin Bandopadhyay**, Chetan S. Mistry. “Effects of Total Pressure Distribution on Performance of Small Size Counter-Rotating Axial-Flow Fan Stage for Electrical Propulsion.” *ASME Gas Turbine India Conference*, 2019.
1. **Tuhin Bandopadhyay**, Shashank Gandhi, N. K. Peyada, A. Ghosh. “Handheld FFT Analyser.” *7th The International Conference on Theoretical, Applied, Computational and Experimental Mechanics*, 2017.

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant

Fall 2023, Fall 2022, Spring 2022 Semesters

Incompressible flow (AE311)

Aerospace Engineering, UIUC

- Taught course modules, held office hours, helped develop academic instructional materials (HWs, projects), created lab demonstration videos, and graded HWs and projects.

Research mentoring

2021-Present

Undergraduate and graduate students mentor

Aerospace Engineering, UIUC

- Mentored six undergraduate students to perform semester-long research projects in our research laboratory.
- Mentored two graduate students toward their Master’s thesis.

TECHNICAL SKILLS

Experimental techniques

PIV, PTV, Hot-wire anemometry, MRV, MRC, pressure, and temperature sensors, load-cell.

Programming

C, C++, Python, Matlab, Labview, Arduino.

Software

Fluent, CFX, ICEM CFD, Turbogrid, Solidworks, Catia, Tecplot.

Operating System

Windows, Linux based operating systems (Ubuntu, Raspbian).